

M PHARM
(SEM I) THEORY EXAMINATION 2022-23
MODERN PHARMACEUTICAL ANALYTICAL TECHNIQUES

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 10 x 2 = 20

- (a) Define membrane indicator electrodes.
- (b) Name the two official compounds which can be assayed by potentiometric titrations.
- (c) What are the three essential requirements for thermogravimetric analysis?
- (d) How does ion chromatography work?
- (e) What do you mean by the term electrophoresis?
- (f) Give any two desorption ionization techniques in mass spectroscopy.
- (g) Define ^{13}C NMR spectroscopy.
- (h) Explain the term spectrofluorimetry.
- (i) What do you understand by fingerprint region in IR spectroscopy?
- (j) Write about affinity chromatography.

SECTION B

2. Attempt any twoparts of the following: 2 x 10 = 20

- (a) Discuss about the principle involved, instrumentation and factors affecting resolution of TLC or HPLC.
- (b) Define chemical shift. How to measure chemical shift? And explain the factors affecting the value of chemical shifts.
- (c) Write in detail about RIA and ELISA.

SECTION C

3. Attempt any fiveparts of the following: 7 x 5 = 35

- (a) Explain the merits of AAS over FES.
- (b) Discuss the instrumentation of paper chromatography.
- (c) Define Bragg's law. How X-Rays are produced, explain with suitable diagram?
- (d) Give a detailed note on gel electrophoresis.
- (e) Discuss about mass fragmentation with the reference of metastable and isotopic peaks.
- (f) Describe spin-spin coupling with suitable example of spectra.
- (g) Write a short note on instrumentation and applications of DTA.